

Raj Tibarewala

AI / ML Engineer · Software Engineer — machine learning for autonomous systems

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EDUCATION **Vellore Institute of Technology, Vellore** Jul 2024 – May 2028
B.Tech Computer Science and Engineering (Cyber Security) · CGPA 8.65 / 10

EXPERIENCE **Team Ardra — Software & R&D, Autonomous Drone Systems** Sep 2025 – Present

Build and validate embedded Linux software for onboard compute (NVIDIA Jetson Orin Nano, Raspberry Pi) on autonomous drone platforms.

Prototyped an onboard vision-serving architecture (FastAPI, PyTorch, Hugging Face DETR) that decouples camera capture, threaded model inference and live video streaming, so slow inference does not stall the feed.

Deploy edge ML inference within latency, power and compute budgets; debug integration issues across sensor, compute and control layers.

Pinnacle Labs — Cybersecurity Intern (remote) Feb – Mar 2026

Built Python security tooling: a hybrid AES + RSA text-encryption tool and an input-capture research tool examining interception techniques and how they are detected.

PROJECTS **AegisFlight — ML-Based Drone Intrusion Detection** 2026 · Ongoing

IIT Bombay TechFest 2026, PUSHPAK Grand Challenge (Stage 1 submitted) · Python, scikit-learn, NumPy, Pandas, pymavlink, FastAPI, React

Explainable IDS over simulated MAVLink 2 telemetry, fusing protocol-rule, physics-consistency, ML anomaly (benign-only Isolation Forest + Mahalanobis over 11 online features) and SHA-256 firmware-integrity detectors.

Reproducible evaluation: session-level splits, multi-seed attack-scenario benchmark (23k+ decisions), ML-off ablation, and replay of public PX4 / ALFA flight logs for false-alarm and sim-to-real (KS test) analysis.

FastAPI / WebSocket backend, React dashboard, and a pytest suite spanning unit, integration, end-to-end and API tests.

GeoAI — Rooftop Mapping from Drone Orthophotos 2025 · Finalist

IIT Bombay TechFest 2025 · Python, Ultralytics YOLOv8, SegFormer, OpenCV, rasterio, Shapely, Docker

Turned village-scale GeoTIFF orthophotos and shapefiles into training data: CRS alignment, 640×640 tiling, empty-tile filtering, shapefile-to-label generation, 70/20/10 split, label and CRS validation, visual QA.

YOLOv8 rooftop detection with four-class roof classification (RCC, Tin, Tiled, Others) and utility detection; SegFormer segmentation of water bodies and roads; Dockerized inference.

Hubble Constant Estimation — Astrophysics Data Analysis 2025

India Space Academy · Python, NumPy, Pandas, Matplotlib

Estimated the Hubble constant and the age of the universe from Type Ia supernova data; validated against Planck 2018 results.

Infinity-Snap — Gesture-Based File Automation 2025

Python, OpenCV, MediaPipe

Real-time computer-vision pipeline mapping webcam-detected hand gestures to OS-level file operations.

RESEARCH **“Nuclear Thermal Rocket: Design Considerations and Efficiency Optimization for Deep-Space Explorations”**

Co-authored under faculty supervision — literature review, background research, technical structuring and manuscript preparation.

SKILLS

Languages	Python, C++, C, Java, JavaScript, SQL
ML & data	scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Jupyter; anomaly detection, feature engineering, model evaluation, ablation, statistical analysis
Deep learning & CV	PyTorch, Ultralytics YOLO, Hugging Face Transformers, OpenCV; object detection, semantic segmentation, geospatial imagery (rasterio, Shapely)
Systems	FastAPI, Docker, Linux, Git / GitHub, pytest, NVIDIA Jetson Orin Nano, Raspberry Pi
Frontend	React, JavaScript
Learning	LLMs, RAG, LangChain, inference optimization